

KAOUSHEIK JAYAKUMAR

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College Park, Maryland - 20740

Education

University of Maryland, College Park

August 2025 – May 2027

MS in Computer Science

Advised by Prof. Dinesh Manocha

Indian Institute of Technology Madras

2018 – 2023

Dual Degree (B.Tech + M.Tech) in Electrical Engineering, Minor in Artificial Intelligence

CGPA: 8.59/10

Advised by Prof. S. Umesh and Prof. Hema Murthy

Publications

- A Closer Look at Failure Modes in Temporal Understanding of Large Audio-Language Models**
Apoorva Kulkarni, **Kaousheik Jayakumar**, Sreyan Ghosh, Sarah Wiegrefe, Dinesh Manocha, Ramani Duraiswami
Under review at ARR (Jan) – San Diego, 2026
- Do Audio-Visual Large Language Models Really See and Hear?**
Ramaneswaran Selvakumar*, **Kaousheik Jayakumar***, S Sakshi, Sreyan Ghosh, Ruohan Gao, Dinesh Manocha
Under review at CVPR – Denver, 2026
- Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents**
Mahesh Ramesh, **Kaousheik Jayakumar**, Aswinkumar Ramkumar, Pavan Thodima, Aniket Rege
Under review at ICML – Seoul, 2026
- Are LLMs Generalist Hanabi Agents?**
Mahesh Ramesh, **Kaousheik Jayakumar**, Aswinkumar Ramkumar, Pavan Thodima
*ICML(W), ICCV(W), NeurIPS(W) * 3 – San Diego, 2025*
- The Tag-Team Approach: Leveraging CLS and Language Tagging for Enhancing Multilingual ASR**
Kaousheik Jayakumar, Vrunda N. Sukhadia, A Arunkumar, S Umesh
Interspeech – Dublin, 2023
- Building Robust and Scalable Multilingual ASR for Indian Languages**
Arjun Gangwar, **Kaousheik Jayakumar**, S Umesh
Arxiv preprint - 2024

Work Experience

Oracle

Bangalore, Karnataka

Member of Technical Staff

July 2023 – Aug 2025

- Integrated edge computing solutions into Oracle Cloud Infrastructure (OCI), unifying architectural components to optimize efficiency, reduce latency, and enhance overall system performance across cloud environments.
- Engineered and executed a large-scale OS upgrade to Oracle Linux 8, ensuring cross-compatibility of dependent components, libraries, and middleware and improved stability of rack-mounted infrastructure.
- Engineered a solution to partition large ISO images into smaller chunks for parallelized security scanning, integrating it back into the infrastructure while ensuring compatibility with existing components, storage, and network pipelines.

Oracle

Bangalore, Karnataka

Product Development Intern

May 2022 – July 2022

- Optimized Docker images by implementing multi-stage builds and minimizing layers, reducing image size and improving deployment speed and runtime efficiency.
- Developed an automated CICD pipeline system that integrated multiple components to generate RPMs and ISO builds
- Automated the configuration management of build environments using Bash/Python scripts, eliminating manual setup overhead and ensuring consistency across development and staging.

- Developed an internal tool to detect vulnerabilities on open ports of any ASN / Organization
- Enhanced machine learning algorithms that were automated into cyber-security solutions for detecting anomalies

Projects and Research Experience

Mitigating Hallucinations in Omni Models via Audio-Visual Semantic Confidence |

- Developed Audio-Visual Semantic Confidence method achieving 7.6% improvement on MUSIC-AVQA, outperforming five CVPR 2024 baselines with training-free approach.
- Engineered cross-modal attention guidance for 7B-parameter multimodal LLMs, reducing hallucinations by 28% across 18,000 samples with FlashAttention-compatible implementation.

Interpreting the modality bias in Audio-Visual Language models | *CVPR 2026*

- Conducted mechanistic interpretability analysis of Audio-Visual LLMs, revealing 56% performance degradation on counterfactual audio-visual inputs and identifying systematic visual bias in cross-modal information processing through attention pattern analysis.
- Developed novel evaluation framework using counterfactual audio-visual samples and LLM-based judges, demonstrating that models encode rich audio semantics (61% latent understanding) but fail to surface them in generation.
- Performed causal mediation experiments via attention knockouts across transformer layers, proving that blocking visual pathways in deeper layers recovers 50% audio understanding by eliminating cross-modal interference.

Enhancing ASR systems for low-resource languages | *Dual Degree Thesis, IIT Madras*

- Exploited phonemic similarities among various Indian languages and enhanced the performance of multilingual ASR systems for 10 low-resource languages, reducing WER by **2-15%** over the corresponding monolingual baselines.
- Analysed the effect of introducing language ID as an additional learning task which showed further **2-5%** improvement.

National Language Translation Mission (Speech & ASR) | *Bhashini, Govt. of India*

- Developed multilingual Speech systems (ASR, Speech Translation, Synthesis) across multiple Indian languages.
- Curated **4,000 hours** of labelled data from NPTEL lectures for low-resource Indian languages by audio-text alignment.
- Deployed real-time code-switched streaming models for ASR, optimizing inference latency for live-transcription.

Video-to-Video translation pipeline with Lip-Syncing | *IIT Madras*

- Engineered a video-to-video translation pipeline to synchronize lip movements of lecturers with translated audio tracks.
- Implemented and fine-tuned generative models (Wav2Lip/GANs) on technical lecture videos, ensuring temporal consistency and realistic mouth articulation.
- Addressed challenges in pose preservation and face restoration, integrating face detection landmarks to maintain visual fidelity.

Bridge - Multiplayer Co-op Card Game | *Interactive Web-Application*

- Developed a real-time multiplayer application for the classic card game Bridge using React, NodeJs and Socket.IO.

Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, MERN full-stack

Libraries and Frameworks: PyTorch, Tensorflow, Keras, Numpy, Pandas, NodeJS, Django, React, Angular, React Native, NextJS, GraphQL

Databases: Postgres, MySQL, MongoDB

Software and Toolkits: Git, Linux, Shell / Bash scripting, REST API, Docker, Kubernetes, LaTeX, ESPNet, Fairseq

Relevant Coursework: Multimodal Foundational Models, Natural Language Processing (NLP), Probability, Statistics, Computer Vision, Linear Algebra, Data Structures and Algorithms, Estimation Theory, Multi-Arm Bandits

Miscellaneous

Teaching Assistant - Signals and Systems, IIT Madras

Jan 2023 – May 2023

Teaching Assistant - Digital Signal Processing, IIT Madras

July 2022 – Nov 2022

ICPC Regionalist - Secured 120th position out of 721 teams in the Asia Amritapuri regionals

Sept 2020